



Turn watching into remembering.

HHCC 2026 · Iteration Track · Education

ClipQuest turns any public, captioned YouTube lesson into an evidence-grounded quiz, then turns the result into a review habit.

JUDGE FIRST

<https://clipquest.ccwu.cc>

Chrome + the bundled extension + a learner-owned DeepSeek key

- LIVE PRODUCT
- EDUCATION
- ITERATION

Submission	Details
Team	404-Waifu-Not-Found · @404-Waifu-Not-Found/cos
Members	UnoxyRich · JimmyfaQwQ · ILikeLayla · Justin-Yonardo
Repository	github.com/404-Waifu-Not-Found/HHCC2026
Judge path	Open the product, install the supplied extension, add a DeepSeek key, paste a public captioned lesson, and start a quest.

Iteration-track baseline

One week before the event, ClipQuest was a short README and product idea. No frontend, quiz engine, persistence, extension, native path, deployment, or test suite existed. The working product was built during the 36-hour challenge.

36h

BUILD WINDOW

4

RUNTIME SURFACES

10/10

REACHED Q1 IN QA

0

SERVER INFERENCE COST

1. The problem and the thesis

The problem judges should evaluate

Video makes learning feel easy because recognition replaces recall. A learner can finish a lesson, understand it while watching, and still fail to reproduce the idea tomorrow. The usual fixes add friction: notes split attention, flashcards require authoring, and generic AI quiz tools can ask about a topic rather than the lesson actually watched.

Our thesis

The best next step after a lesson is low-friction retrieval practice: ask the learner to produce an answer, correct the misconception immediately, and schedule a return visit.

What ClipQuest changes

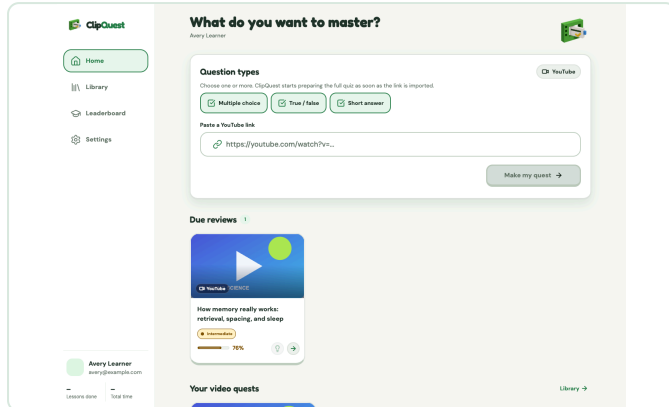
- 1. **Bring your own lesson.** Paste a public YouTube URL and confirm the detected video.
- 2. **Practice, do not just consume.** Choose 5, 10, or 15 questions across multiple choice, true/false, and short answer.
- 3. **Start immediately.** Question 1 is generated and validated first; the rest arrive while the learner answers.
- 4. **Learn from misses.** Every response gets a correction and “Why” explanation. Missed concepts return as adaptive retries.
- 5. **Keep the progress.** Completion shows score, first-try accuracy, missed concepts, mastery, and a one-page cheat sheet. The Library schedules review.

Learning science mapped to product decisions

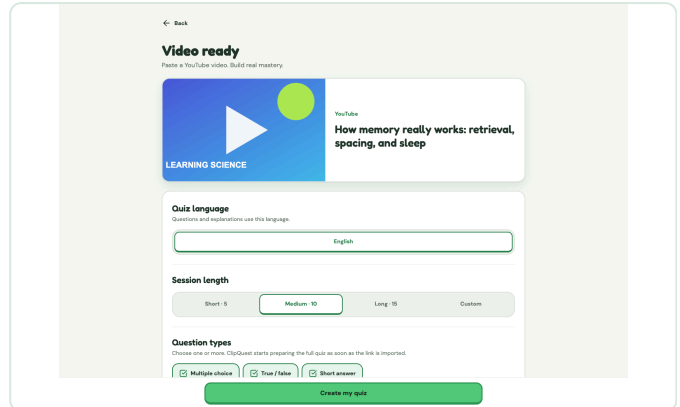
Mechanism	Visible implementation	Judge signal
Retrieval practice	Every lesson becomes an answerable quest.	The learner must produce, not re-watch.
Immediate feedback	Correct answer plus reasoning after each submission.	Misconceptions are corrected in context.
Error-focused review	“What to review” recap and reformulated retries.	Review time targets the learner’s gaps.
Spacing	Library due state and mastery thresholds.	The lesson returns after the session.
Grounding	Questions derive only from usable captions.	Captionless or invalid content fails closed.

Judge lens: does the product convert passive video consumption into a repeatable learning loop?

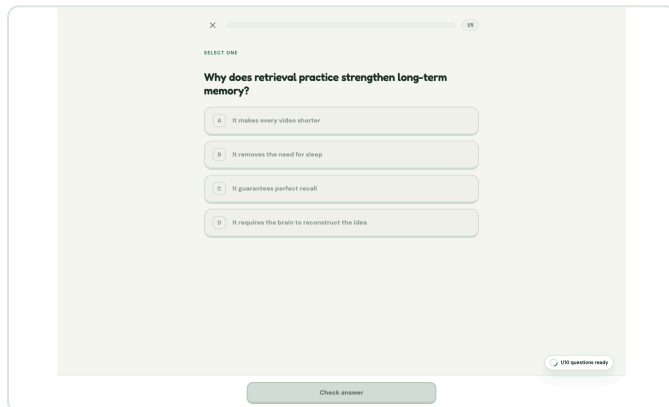
2. The learner journey



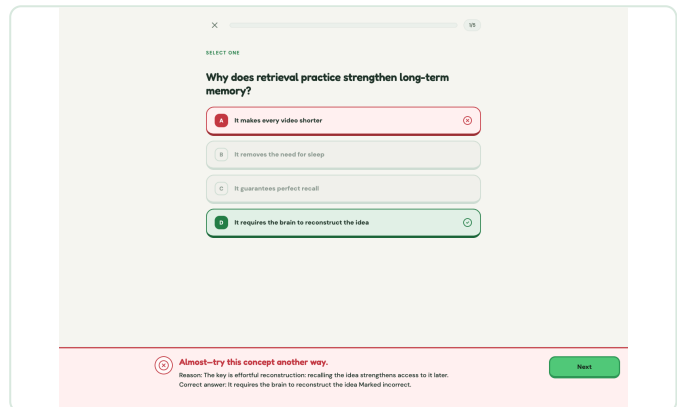
1. Paste a lesson and validate the source.



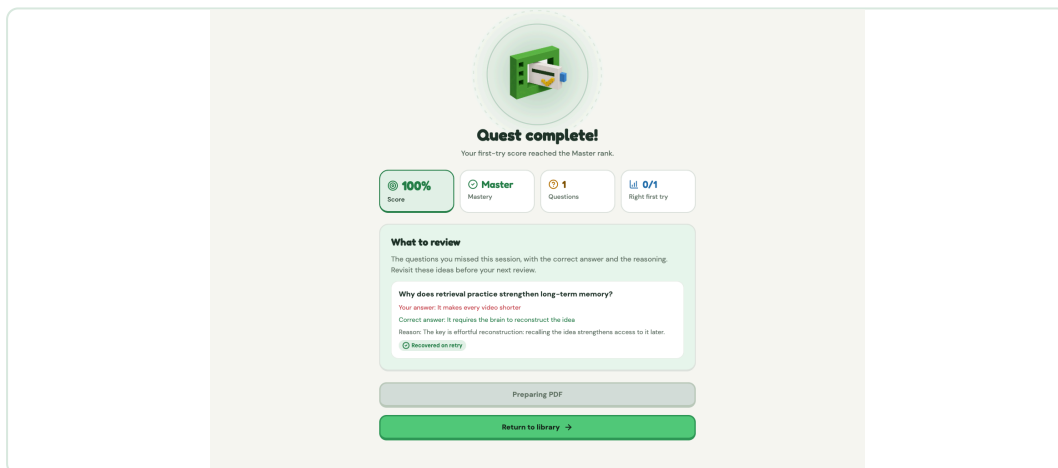
2. Confirm the video and question plan.



3. Start with question 1 while generation continues.



4. See the correction and reasoning immediately.



5. Finish with a targeted recap, mastery state, and a downloadable cheat sheet.

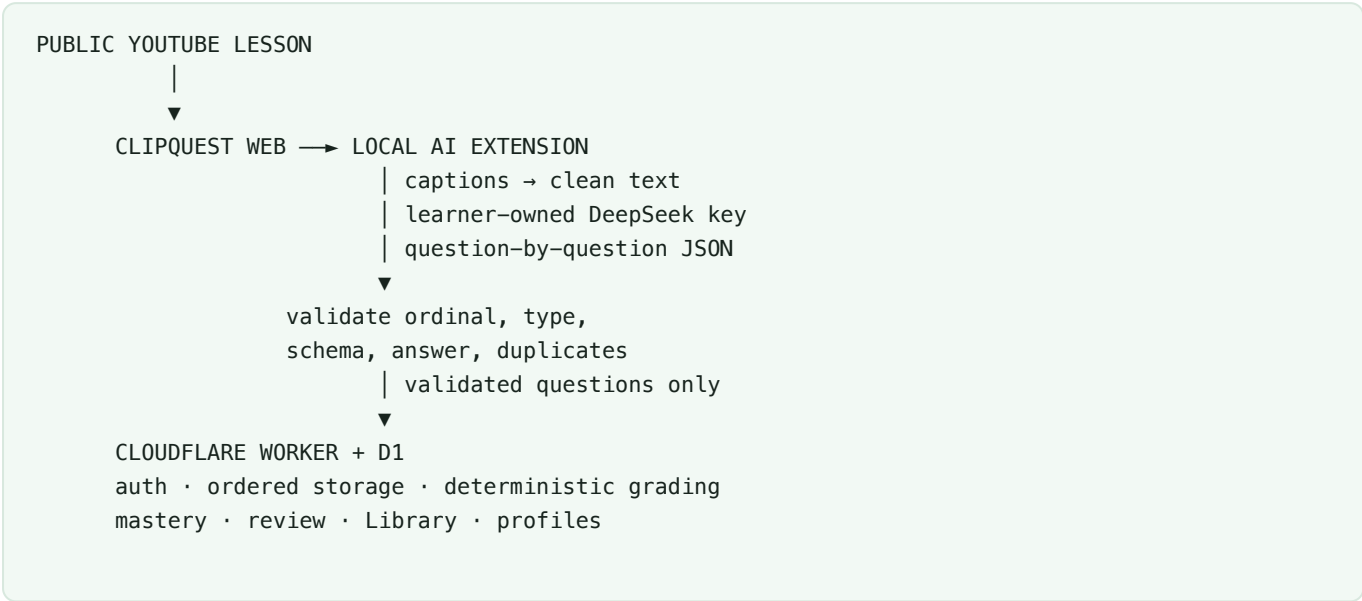
Two-minute judge walkthrough

Use any public lesson with text captions. Make one wrong answer on purpose: the feedback, retry behavior, recap, and Library state are the clearest proof that this is a learning product rather than a quiz generator.

Judge lens: observe time-to-first-question, feedback quality, and whether the next action is obvious.

3. The privacy and reliability boundary

The central design choice is where generation runs. ClipQuest Local AI is a Chrome extension: captions are acquired and normalized in the learner’s browser, then sent directly to DeepSeek with the learner’s own key. The Cloudflare Worker never receives the key, captions, prompts, or raw model output.



Why the boundary matters

- **Privacy:** lesson text and credentials stay outside the server.
- **Economics:** inference cost does not scale with learner count.
- **Correctness:** the Worker cannot invent a fallback because it has no generation path.

Failure-safe generation

Each question is parsed and validated independently. A malformed question 9 cannot erase accepted questions 1–8. Recovery regenerates only the first missing ordinal with bounded retries. If repair fails, the accepted prefix remains and the learner sees an explicit failure state — never a fake question.

Boundary check	Guarantee
Ordinal and requested type	Questions arrive in the selected plan and exact order.
Schema and answer mapping	Malformed objects are rejected before storage.
Duplicate prompt detection	Repeated questions do not silently enter a bank.
Short-answer rubric	Server grading is deterministic and reproducible.

Judge lens: the safety guarantee is structural, not a promise that a model will always behave.

4. What was built in 36 hours

The iteration is measurable because the starting point was a written idea. The team moved from risk to working proof rather than polishing a pre-existing product.

Stage	Risk addressed	Shipped proof
1 • Core loop	Can a learner finish a quest?	Import → answer → grade → completion.
2 • Grounding	Will questions reflect the lesson?	Caption-only acquisition and evidence-bound prompts.
3 • Correctness	What if model output is malformed?	Typed contracts, validation, ordered imports, bounded repair.
4 • Speed	Will waiting kill the habit?	Question 1 opens first; remaining questions stream behind.
5 • Retention	What happens after the score?	Retries, recap, mastery, Library, spaced review, cheat sheet.
6 • Ship	Can judges use it?	Cloudflare deployment, extension package, responsive UI, CI and QA evidence.

Before → after

Before the event	Submitted product
README-only concept	Live web app at clipquest.ccwu.cc
No AI or quiz engine	Local caption-grounded generation with three formats
No persistence	Accounts, attempts, mastery, Library, profiles
No client surfaces	Web, Chrome bridge, PWA, native implementation paths
No quality gate	Contract, API, app, engine, extension, UI, build and asset checks

Team collaboration

UnoxyRich led scope, integration, release engineering, and verification. JimmyfaQwQ owned the extension, captions, generation, validation, and feedback. ILikeLayla owned the learner journey, visual system, and accessibility. Justin-Yonardo owned the Hono API, D1 schema, authentication, and automated QA.

Judge lens: reward the risk removed at each stage, not only the amount of code produced.

5. Technical depth and verification

Layer	Technology	Role
Web	React 19 · TypeScript	Import, quiz, feedback, completion, Library, profiles, themes.
Local AI	Chrome Manifest V3 · JavaScript	Caption access, DeepSeek calls, streaming, validation and repair.
Edge	Cloudflare Workers · Hono · Better Auth	Sessions, ordered storage, deterministic grading and scheduling.
Data	D1 · KV · private R2	Progress, limits, thumbnails, avatars and private PDFs.
Contracts	TypeScript · Zod	One versioned schema across extension, page and Worker.
Quality	Vitest · Node · Playwright · ESLint · Prettier	Unit, contract, API, app, browser, type, lint and build checks.

What is verified

- Ten out of ten tested videos reached question 1 in the current live matrix.
- Deterministic grading checks passed for the accepted answer set; no invented fallback questions were observed.
- Questions are stored only after schema, order, type, duplicate, and answer-consistency checks.
- Production QA artifacts and the current release notes are committed in `docs/`.

Honest limitation

Full-bank completion on the first try reached 5/10 videos in the latest matrix and 8/10 after one retry. The known mismatch failure fails closed, preserving accepted questions instead of substituting content. The in-development Workplace AI chat route is hidden rather than presented as finished.

How to verify the claim

Read the current matrix in `docs/QA-HEADLESS-MATRIX-2026-08-22.md`, inspect the shared contracts in `packages/contracts/`, and follow the generation boundary through `apps/extension/`, `packages/local-quiz-engine/`, and `apps/api/`.

6. Scoring map and next proof

HHCC dimension	Evidence in ClipQuest
Creativity & originality · 30%	A privacy-first browser generation boundary, caption-grounded questions, progressive opening, and an original visual system.
Technical complexity · 30%	Four runtime surfaces joined by typed contracts, streaming protocol, strict ordinal validation, recovery, and storage-only edge architecture.
Functionality & usability · 20%	Live end-to-end quest flow, immediate feedback, adaptive retries, recap, mastery and Library.
Branding & social impact · 10%	Approachable learning identity and a zero-server-inference-cost model for open educational video.
Teamwork & collaboration · 10%	Clear ownership, documented iteration decisions, shared contracts, and joint production verification.

What comes next

1. **Sample lesson:** let visitors experience the loop before installing an extension.
2. **Expanding review:** move from 1 → 3 → 7 → 21 days and schedule by concept.
3. **Teacher view:** turn shared quests into a lightweight class feedback loop.
4. **More evidence:** add caption-moment citations and multilingual generation.

Recommended judging route

Start at the live product. Then inspect one generated question, one incorrect-answer explanation, the completion recap, and the Library. Finally, use the architecture page and QA report to validate why the privacy and no-fallback claims are credible.

Links

Live: <https://clipquest.ccwu.cc>

Code: <https://github.com/404-Waifu-Not-Found/HHCC2026>

Judge README: [README.md#judge-briefing](#)

Documentation: [docs/HACKATHON.md](#) · [docs/README.md](#)

AI usage disclosure: the team used GitHub Copilot and Claude Code CLI for boilerplate, refactoring, documentation, and debugging assistance. The team owned the concept, scope, architecture, privacy boundary, algorithms, testing strategy, and visual direction.